

Efficient Multi-Task Safety Guardrail: Detecting Harmful Content and Jailbreak Attempts in a Single Classifier

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Abstract

Securing Large Language Models (LLMs) from malicious inputs is a key challenge as they grow increasingly ubiquitous in user-facing applications. Most existing defenses use independent classifiers for different threat classes, such as explicit harmful content and adversarial jailbreaks, leading to high computational overhead and latency. This paper presents an Efficient Multi-Task Safety Guardrail framework designed to identify both direct policy violations and advanced adversarial attempts in a single forward pass. By conducting an extensive Exploratory Data Analysis (EDA) on the *WildGuardMix* dataset, we identify critical data quality constraints, including high duplicate volumes and low vocabulary diversity, which we address through rigorous preprocessing. We evaluate a spectrum of classical and deep learning models, identifying Gated Recurrent Units (GRUs) as the optimal lightweight architecture, achieving 81.70% test accuracy and a 0.8100 Macro F1 score. Ultimately, this approach establishes a scalable, low-latency security layer capable of ensuring safe human-AI interactions in resource-constrained deployment environments.

1 Introduction

The rapid integration of Large Language Models (LLMs) across diverse sectors has exposed significant security vulnerabilities, particularly their susceptibility to generating inappropriate or harmful content when subjected to adversarial jailbreak attacks (Yuan et al., 2024). To mitigate these risks, content moderation systems and guardrails have become an industry standard. However, state-of-the-art systems like Llama Guard (Han et al., 2024) often rely on massive, computationally heavy filters or require multiple model inferences, introducing unacceptable latency bottlenecks during live deployment.

Recent literature highlights the necessity of shifting toward highly efficient, multi-task models. Approaches like Free Jailbreak Detection (FJD) (Chen et al., 2025) and RigorLLM (Yuan et al., 2024) propose utilizing the target model’s internal logits or hybrid K-Nearest Neighbor fusions to bypass the need for a secondary LLM classifier. Building upon these principles, this project investigates the development of a unified, lightweight classifier. We utilize the *WildGuardMix* dataset (Han et al., 2024) to train traditional machine learning and recurrent neural network (RNN) architectures, aiming to match the safety alignment goals of comprehensive frameworks like Aegis2.0 (Ghosh et al., 2025) and BeaverTails (Ji et al., 2023) without the associated computational burden.

2 Exploratory Data Analysis

To construct a robust multi-task classifier, we first analyzed the *WildGuardMix* dataset, which categorizes inputs into four distinct classes: Benign_Vanilla, Benign_Adversarial, Harmful_Vanilla, and Harmful_Adversarial.

2.1 Dataset Dimensions and Quality

The initial dataset consisted of 86,759 training samples and 1,699 test samples. Our EDA revealed several critical data quality anomalies that pose a high risk of model overfitting or data leakage:

- **Empty and Short Prompts:** The training set contained 14 completely empty prompts and numerous single-word outliers (e.g., “okay”, “Yes.”).
- **Massive Duplication:** A total of 38,907 duplicate prompts were identified in the training set, necessitating strict deduplication before model training.

- **Vocabulary Diversity:** The vocabulary diversity ratio was extremely low at 0.0186 (127,078 unique tokens out of 6,847,857 total tokens), indicating a highly repetitive linguistic structure across the adversarial and vanilla prompts.

2.2 Text Statistics

Following the removal of stopwords and punctuation, we observed that the average prompt length is approximately 79 words. However, the data exhibits extreme right-skewness, with the maximum prompt reaching 1,960 words (12,814 characters). Boxplot distributions confirmed that adversarial prompts (Benign_Adversarial and Harmful_Adversarial) consistently possessed higher word and character counts compared to their vanilla counterparts, highlighting the verbose nature of roleplay-based jailbreak attacks.

3 Methodology

Our methodology contrasts classical bag-of-words machine learning models against deep sequential architectures capable of capturing the long-range dependencies inherent in adversarial attacks.

3.1 Feature Engineering

Two primary feature extraction pipelines were deployed:

1. **TF-IDF Vectorization:** Applied for classical ML models, utilizing unigrams and bigrams with a maximum feature limit of 8,000 to balance signal and noise.
2. **Pretrained GloVe Embeddings:** Used for deep learning models. Sequences were padded to the 98th-percentile token length, and a 100-dimensional GloVe matrix (glove.6B.100d) was mapped to a vocabulary cutoff capturing 98% of all token occurrences.

3.2 Model Architectures

We evaluated nine distinct architectures. The classical baselines included Multinomial Naive Bayes, Logistic Regression (L1/L2 regularized), and Random Forest. The deep learning suite explored standard SimpleRNN, Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) models, alongside their bidirectional counterparts.

4 Results

The models were evaluated on the held-out test set using Accuracy and Macro F1 scores to account for any residual class imbalances. The final comparative leaderboard is presented in Table 1.

Rank	Model	Accuracy	Macro F1
1	GRU	81.70%	0.8100
2	Bi-GRU	80.69%	0.8001
3	LSTM	80.22%	0.7964
4	Logistic Regression	79.34%	0.7892
5	Bi-LSTM	79.11%	0.7852
6	SimpleRNN	78.46%	0.7780
7	Bi-SimpleRNN	76.75%	0.7627
8	Random Forest	73.16%	0.7282
9	Naive Bayes	69.45%	0.6926

Table 1: Final Test Set Comparison Leaderboard.

5 Discussion

The results demonstrate a clear hierarchy in architectural suitability for multi-task safety classification. The **GRU** emerged as the best overall performer. Its success can be directly tied to the EDA findings: adversarial jailbreaks often disguise harmful intent within lengthy, complex phrasing. The GRU’s gating mechanism efficiently retains these long-range sequential dependencies without succumbing to the vanishing gradient problems seen in the SimpleRNN.

Conversely, the **Naive Bayes** classifier performed the worst (69.45% Accuracy). As a probabilistic bag-of-words model, it entirely discards word order, making it highly susceptible to the adversarial wrappers that pad harmful requests with benign vocabulary. **Logistic Regression** performed admirably (79.34%) by leveraging TF-IDF bigrams, but still fell short of the recurrent models’ ability to synthesize sequence context.

6 Conclusion

This study highlights the viability of lightweight, sequential models for LLM content moderation. By rigorously cleaning the *WildGuardMix* dataset and deploying a GRU architecture, we achieved highly accurate, single-pass classification of both overt harms and covert jailbreaks. Future work will explore integrating inference-time logit inspection techniques (Chen et al., 2025) alongside this recurrent baseline to further optimize computational efficiency.

References

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